Chen 2020 NPC — recapitulation summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → **NO-GO**

Continuous (1 connected component) but NOT sample-mixed: ~50% of cells in patient-pure clusters at θ=2 (≈90% at θ=1), robust across n_pcs ∈ {15,20,30} — not the attempt-1 n_pcs=6 artifact. Purest clusters = non-cycling P11 (patient-private CNV). Only θ=3 mixes, by over-integrating.

Gate 2 — module geography → **NOT REACHED (NA)**

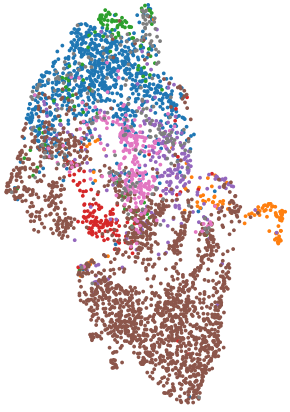
Not evaluated — no sample-integrated continuous manifold on which to fit a trajectory / score module geography.

Gate 3 — hub mutual exclusivity → **NOT REACHED (NA)**

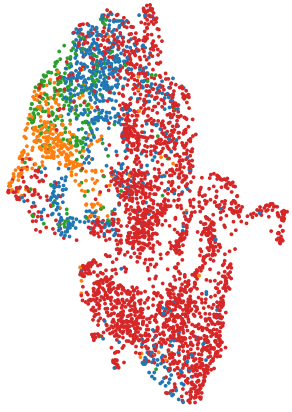
Not evaluated — requires a branching trajectory with an identified hub, which Gate 1 precludes.

STEP FIGURES

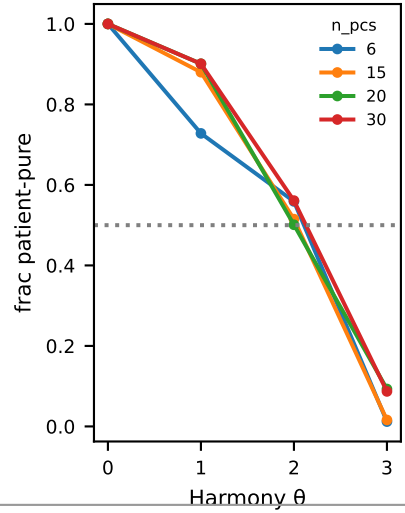
Patient (8 kept)



Cell cycle



Gate 1 sweep



PRRX1 (62% detect)



Dataset. Nasopharyngeal carcinoma (Chen et al., 2020), 10x — 14 patients / 71,896 cells (master scope); malignant only: 3,763 load → 3,674 across 8 patients (≥62 cells) enter the embedding.

Verdict. ✗ Pattern does NOT reproduce — Gate 1 NO-GO. The malignant compartment is patient-segregated (inter-patient CNV heterogeneity), not a shared sample-integrated developmental manifold. No hub, no trajectory, no cross-repression axis to test. EMT-TFs shown for reference: PRRX1 / SNAI2 / FOXC1 (FOXC/PRRX families; canonical SNAI1 & TWIST1 fall below the 3% detection floor).

Parameter choices

Min cells/sample	≥ 62 (median-restricted gap; keeps 8/14)	**Embedding**	UMAP, Leiden res=0.5 → 9 clusters
Genes	1,290 over-dispersed (GAM)	**Trajectory**	none (Gate 1 NO-GO)
n_pcs	20 (re-derived; Kürten 20 ≤ n ≤ Puram 40)	**EMT-TF panel**	detection-gated ≥ 3%; PRRX1/SNAI2/FOXC1
Integration	Harmony on sample, θ=2, 20 it	**Geometry**	patient-segregated (not a manifold)